REDBIRD FLIGHT SIMULATIONS

QUALIFICATION AND APPROVAL GUIDE (QAG)

LD, SD, FMX, MCX ADVANCED AVIATION TRAINING DEVICES
AMENDED QUALIFICATION AND APPROVAL GUIDE
VERSION 4.6E



Advanced Aviation Training Device(s)

2301 East St. Elmo Rd., Suite 100
Austin, Texas 78744
(512) 301-0718 Phone
(512) 301-0770 Fax
info@redbirdflight.com

February 15, 2019

FAA APPROVED QAG
Signature and Date

SHAWN M HAYES
Digitally signed by SHAWN M HAYES
Date: 2019.02.26 14:07:17 -05'00'

Shawn Hayes
Manager, Airmen Training and Certification Branch

Table of Contents

LOG OF REVISIONS 5

SECTION 1: COMPLIANCE STATEMENT 6

SECTION 2: AVIATION TRAINING DEVICE (ATD) DESCRIPTION AND PICTURES 7

- Redbird LD 8
- Redbird SD 8
- Redbird FMX 8
- Redbird MCX/LD 9
- Redbird MCX/SD 9
- Redbird MCX 9
- Configuration Components 12
  - Instructor's Station 12
  - Aircraft Instrument Configurations 12
  - Modular Controls 13

SECTION 3: TRAINING DEVICE COMPONENTS LIST 17

- Statements of Compatibility of Software and Hardware 19

SECTION 4: AVIATION TRAINING DEVICE (ATD) DESIGN CRITERIA LIST 20

- Basic ATD Requirements List [Appendix B items] 20
- Advanced ATD Requirements List [Appendix C items] 24

SECTION 5: AIRCRAFT CONFIGURATIONS 25

- List of Previously Approved Configurations (QAG v4.5E) 25
  - Beechcraft BE36-S1 25
  - Beechcraft BE36-G1 26
  - Beechcraft BE55-S1 27
  - Beechcraft BE58-S1 28
  - Beechcraft BE58-S2 29
  - Beechcraft BE58-G1 30
  - Beechcraft BE76-S1 31
  - Cessna C172-S1 32
  - Cessna C172-S2 33
  - Cessna C172-G1 34
  - Cessna C172-G2 35
  - Cessna C182-S1 36
  - Cessna C182-S2 37

Cessna C182-G1 38
Cessna C182-G2 39
Cessna C206-S1 40
Cessna C206-S2 41
Cessna C206-G1 42
Cessna C206-G2 43
Cessna C206T-S1 44
Cirrus SR20-G1 45
Cirrus SR20-G2 46
Cirrus SR22-G1 47
Cirrus SR22-G2 48
Diamond DA20-S1 49
Diamond DA40-G1 50
Diamond DA40-G2 51
Diamond DA42-G1 52
Diamond DA42L-G2 53
Liberty LXL2-S1 54
Liberty LXL2-S2 55
Mooney M20-G1 56
Piper Warrior PA28-60-S1 57
Piper Warrior PA28-60-S2 58
Piper Warrior PA28-60-G1 59
Piper Archer PA28-80-S1 60
Piper Archer PA28-80-S2 61
Piper Archer PA28-80-G1 62
Piper Arrow PA28-2R-S1 63
Piper Arrow PA28-2R-S2 64
Piper Arrow PA28-2R-G1 65
Piper Arrow PA28-2R-G2 66
Piper Seneca PA34-S1 67
Piper Seneca PA34-G1 68
Piper Seminole PA44-S1 69
Piper Seminole PA44-G1 70
Piper Seminole PA44T-S1 71

Socata TB10-S1 72
Tecnam P2006-G1 73
Additional configurations included in this version (QAG v4.6D) 74
Piper Seneca PA34T-S1 74
Piper Seneca PA34T-G1 75
Performance Table 76
SECTION 6: VISUAL SYSTEM WITH IFR, VFR, DAY, AND NIGHT CAPABILITY 78
SECTION 7: ATD FUNCTIONS & MANEUVERS CHECKLISTS 83

LOG OF REVISIONS

Description of Changes	Version	Affected Pages	Edited By
09/19/2017 Submitted Version	4.5	All	WJ
Added Log of Revisions	4.5a	4	WJ
Renamed Section 1 Compliance Statement	4.5a	5	WJ
Moved Group 1 from Section 2 to Section 5	4.5a	5 – 21	WJ
Added individual simulator descriptions	4.5a	3 – 5	WJ
Consolidated Groups 2 – 5 into Configuration Components	4.5a	-	WJ
Moved Figures 5.4 – 5.7 to Section 6	4.5a	49, 50	WJ
Deleted Figured 5.8 – 5.10	4.5a		WJ
Moved and reformatted Performance Table to Section 5	4.5a	45,46	WJ
Section 4 rewrite	4.5a	15 – 19	WJ
Added W&B, component list to Section 5 aircraft configurations	4.5a	20 – 44	WJ
Corrected typographical error in DA42L W&B	4.5a	43	WJ
Updated Images in Section 2	4.5B	11,12	WJ
Updated images in Section 5	4.5B	21-69	WJ
Added Figures 4.7 – 4.17 to Section 2	4.5C	11-13	WJ
Added Control Loading (Single) and (Dual) yokes to Section 3	4.5C	15	WJ
Added Control Loading (Single) and (Dual) rudder pedals to Section 3	4.5C	15	WJ
Added Figures 6.2 and 6.2 Optional CAPS to Section 2	4.5C	11	WJ
Added Optional CAPS to Section 3	4.5C	15	WJ
Updated panel images – Section 5	4.5D	22-70	WJ
Added image 7 & 8	4.5E	8	WJ
Updated image 5.2	4.5E	9	WJ
Updated GLS approach to “No” in Section 7	4.5E	78	WJ
Updated for AC-136B Compliance	4.6	6, 8, 18-23	WJ
Added reference to Figures 5.8 and 5.9 for optional equipment	4.6	8	WJ
Added 27” (6) and (8) monitor options to Component List	4.6	17	WJ
Moved configurations added in 4.5E to Previously Approved	4.6	24 – 73	WJ
Updated description to include dimensions for larger monitors.	4.6	76	WJ
Added Figures 5.8 and 5.9	4.6	79	WJ
Updated text in Compliance Statement	4.6A	6	WJ
Listed SD and MCX/SD as individual models	4.6A	8, 9	WJ
Enlarged image for Figure 5.2	4.6A	12	WJ
Added avionics manufacturer information in Component List	4.6A	17	WJ
Updated text for C3.1.9	4.6A	24	WJ
Updated Procedures and Tasks Test Checklist	4.6A	82-85	WJ
Added description for MCX/LD	4.6B	9	WJ
Added PA34T configurations	4.6C	74-76	WJ
Updated Figure descriptions to include monitor size	4.6D	80, 81	WJ
Corrected typo in Section 7 on item 5)	4.6E	86	WJ

SECTION 1: COMPLIANCE STATEMENT

This Qualification and Approval Guide (QAG) includes a detailed description of all components, functions, and capabilities for the LD, SD, FMX, MCX Redbird models. This includes any optional aircraft configurations with quality color pictures and diagrams. The required information as described in the current advisory circular, AC 61-136B / FAA Approval of Aviation Training Devices (ATD) and Their Use for Training and Experience, required for FAA approval, is contained within this document. This includes listing all the required qualifying items, functions, and capabilities. This QAG is used to clearly describe and verify the functionality of this aviation training device platform confirming its suitability for airman training and experience. A valid FAA Letter of Authorization (LOA) specifying the ATD allowances must also accompany the training device when utilized for airman training or experience requirements as specified in 14 CFR §61 or 141.

This ATD must continue to maintain its performance and function without degradation. The operator of this device is expected to maintain its condition and functionally when used for airmen training or experience requirements. Only the configurations approved for this model, as described within this QAG, can be utilized when satisfying FAA experience or training requirements. Any additions, changes, or revisions to the model or to the available configurations described within this document must be approved in writing by The General Aviation and Commercial Division, AFS-800.

Any questions concerning FAA approval of ATDs should be directed to the General Aviation and Commercial Division, AFS-800.

SECTION 2: AVIATION TRAINING DEVICE (ATD) DESCRIPTION AND PICTURES

The Redbird models are based on the dimensions and layout of a several production aircraft. These models closely represent the overall functionality, performance, avionics, and instrumentation. The platform consists of a cockpit section, instructor's control station, visual display system and an audio system. It incorporates a combination of hardware and software components that is assembled and checked by Redbird Flight Simulations. All hardware elements are permanently installed and designed so the cockpit has the appearance and feel of an actual aircraft. From the pilot's seated position, there are no computer hardware elements such as keyboards, pointing devices, etc. for his or her use.

The Redbird simulators provide realistic and true-to-scale cockpit design, avionics, and reliable hardware/software performance. This platform also provides an effective training environment for student and certificated pilots. This includes the capability of practicing scenario based flight training events, simulated equipment failure and emergency procedures, pilot evaluations, instrument procedures/ experience, and facilitating increased pilot proficiency overall.

The Redbird family of simulators is versatile, expandable and affordable devices that have been designed to be representative of popular single and twin-engine piston aircraft. They offer the user the ability to change the cockpit controls and aircraft models to represent a wide range of popular training aircraft. The line includes the LD, SD, FMX, and MCX models.

Redbird LD

- Wrap-around exterior visuals provided by 6 LCD screens. See Figure 5.4
- Single pilot control with optional 2-axis control-loaded yoke and pilot rudder pedals. See Figures 2.1-2.3 and 5.3 (Optional pilot seating platform required for center-stick yoke)
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 4.1, 4.2
- An interchangeable instrument panel, throttles, and yokes to allow a quick-change configuration options to the end user. See Figures 3.1-3.8
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructor's station, allowing the instructor to operate from inside or outside the simulator. See Figure 5.2

Redbird SD

- Enclosed cockpit with pilot and copilot seating. See Images 2, 7, 8
- Wrap-around exterior visuals provided by 6 or 8 LCD screens. See Figures 5.4, 5.5, 5.8, 5.9
- Single pilot control with optional 2-axis control-loaded yoke and pilot rudder pedals. See Figures 2.1-2.3 and 5.3
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 4.1, 4.2
- An interchangeable instrument panel, throttles, and yokes to allow a quick-change configuration options to the end user.
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructor's station, allowing the instructor to operate from inside or outside the simulator. See Figure 5.2

Redbird FMX

- Enclosed cockpit with pilot and copilot seating. See Images 2, 7, 8
- 3-axis electric motion platform providing pitch, roll and yaw motions. See Images 5,6
- Wrap-around exterior visuals provided by 6 or 8 LCD screens. See Figures 5.4, 5.5, 5.8, 5.9
- Single pilot control with optional 2-axis control-loaded yoke and pilot rudder pedals. See Figures 2.1-2.3 and 5.3
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 4.1, 4.2
- An interchangeable instrument panel, throttles, and yokes to allow a quick-change configuration options to the end user.
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructor's station, allowing the instructor to operate from inside or outside the simulator. See Figure 5.2

Redbird MCX/LD

- Wrap-around exterior visuals provided by 6 LCD screens. See Figure 5.4
- Dual pilot controls with optional 2-axis control-loaded yoke and interconnected rudder pedals. See Figure 2.4 and 5.11
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 4.1, 4.2
- An interchangeable instrument panel, throttles, and yokes to allow a quick-change configuration options to the end user. See Figures 3.1-3.8
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructor's station, allowing the instructor to operate from inside or outside the simulator. See Figure 5.2

Redbird MCX/SD

- Enclosed cockpit with pilot and copilot seating. See Images 2, 7, 8
- Wrap-around exterior visuals provided by 6 or 8 LCD screens. See Figures 5.4, 5.5, 5.8, 5.9
- Dual pilot controls with optional 2-axis control-loaded yoke and interconnected rudder pedals. See Figure 2.4 and 5.11
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 4.4-4.6
- An interchangeable instrument panel and throttles to allow a quick-change configuration options to the end user. See Figures 3.1-3.8
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructor's station, allowing the instructor to operate from inside or outside the simulator. See Figure 5.2

Redbird MCX

- Enclosed cockpit with pilot and copilot seating. See Images 2, 7, 8
- 3-axis electric motion platform providing pitch, roll and yaw motions. See Images 5,6
- Wrap-around exterior visuals provided by 6 or 8 LCD screens. See Figures 5.4, 5.5, 5.8, 5.9
- Dual pilot controls with optional 2-axis control-loaded yoke and interconnected rudder pedals. See Figure 2.4 and 5.11
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 4.4-4.6
- An interchangeable instrument panel and throttles to allow a quick-change configuration options to the end user. See Figures 3.1-3.8
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructor's station, allowing the instructor to operate from inside or outside the simulator. See Figure 5.2



Image 1: Redbird LD



Image 2: Redbird Cockpit Enclosure



Image 3: Redbird FMX and SD Instrument Panel



Image 4: Redbird MCX Instrument Panel

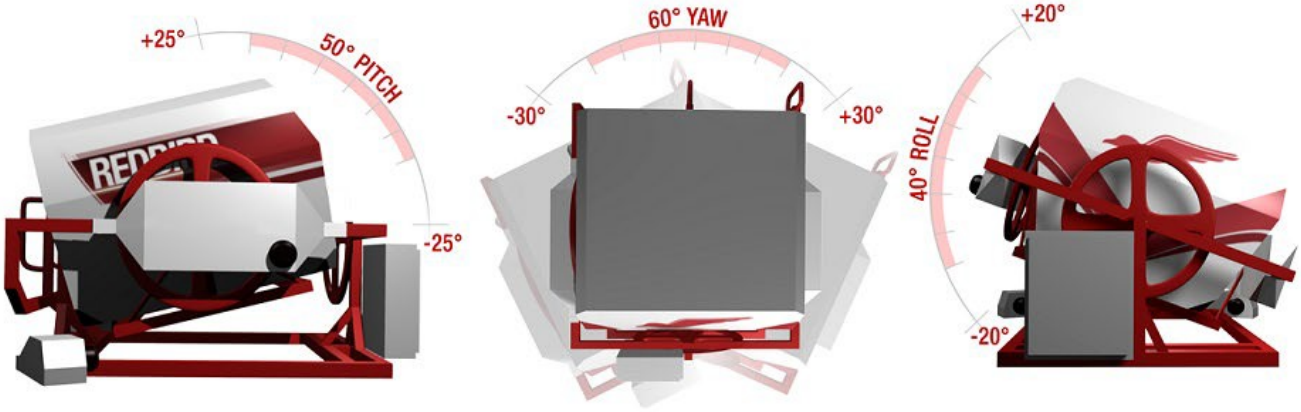


Image 5: Redbird Motion Platform



Image 6: Redbird FMX/MCX Exterior & Motion Platform

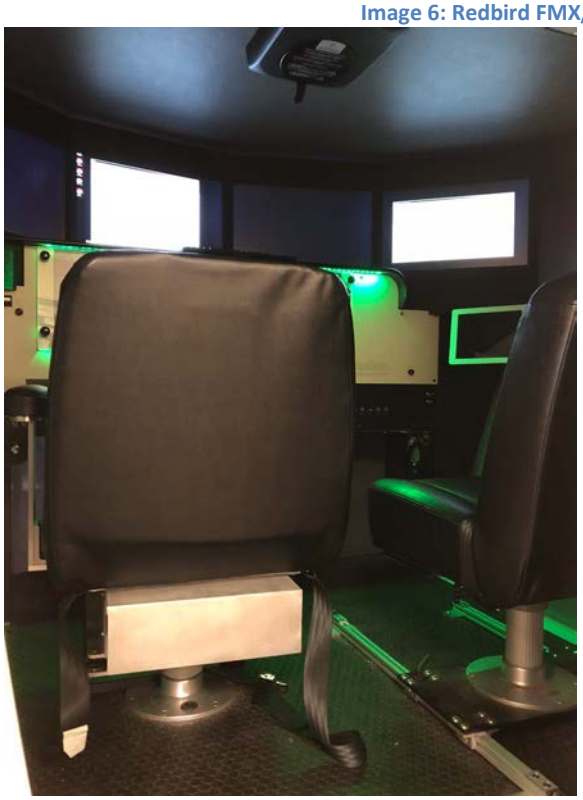


Image 7: Redbird FMX Seats (w/ center stick yoke mount)



Image 8: Redbird MCX Pilot & Co-pilot Seats

Configuration Components

Instructor's Station

The Redbird Instructor Station interface is operated through any PC or browser enabled device.

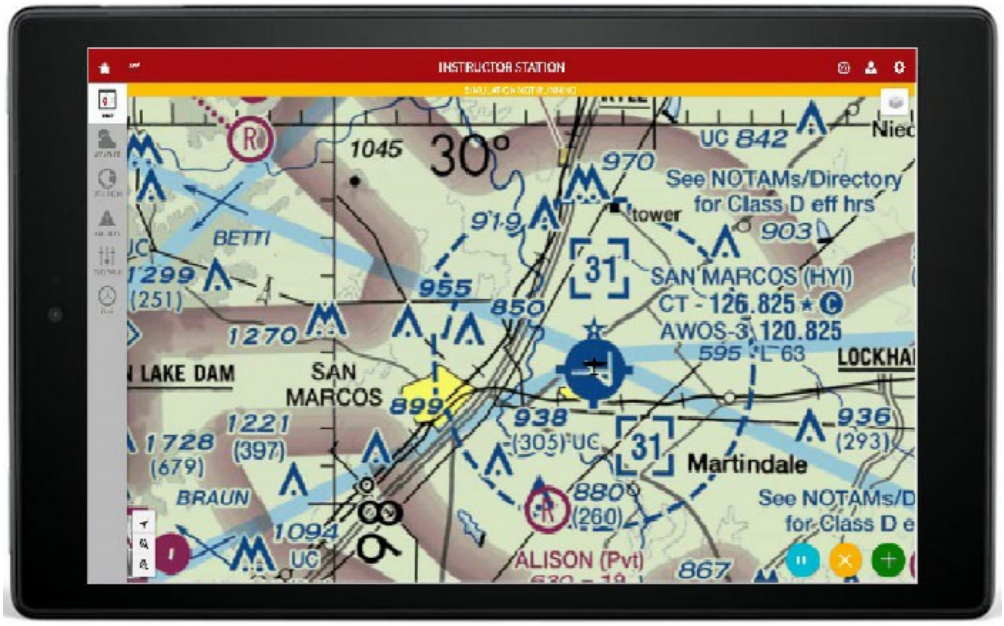


Figure 5.2: Instructor's interface (Mobile Device)

Aircraft Instrument Configurations

The Redbird LD, SD, FMX, and MCX simulators are capable of supporting multiple aircraft configurations utilizing both traditional analog instruments as well as glass panels.



Figure 5.12: Traditional 6-pack Instrument Panel Configuration (Example)



Figure 5.13: Glass Panel Instrument Configuration (Example)

Modular Controls

The modular design of the Redbird LD, SD, FMX, and MCX simulators allow for quick configuration changes between single and multi-engine aircraft configurations.



Figure 2.1: Center Yoke



Figure 2.2: Side Yoke



Figure 2.3: Center Stick



Figure 2.4: Dual Yoke



Figure 3.1: Single Engine Vernier Throttle Quadrant



Figure 3.2: Single Engine Complex Vernier Throttle Quadrant

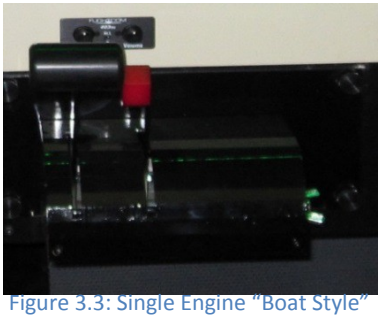


Figure 3.3: Single Engine "Boat Style" Throttle Quadrant

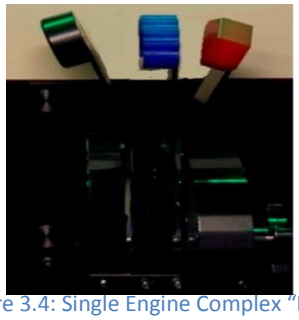


Figure 3.4: Single Engine Complex "Boat Style" Throttle Quadrant



Figure 3.5: Twin Engine Throttle Quadrant – Type 1



Figure 3.6: Twin Engine Throttle Quadrant – Type 2

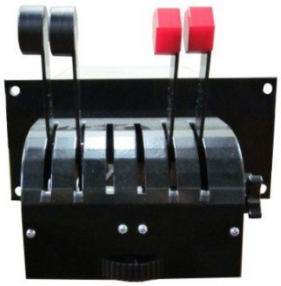


Figure 3.7: Twin Engine Throttle Quadrant – Type 3

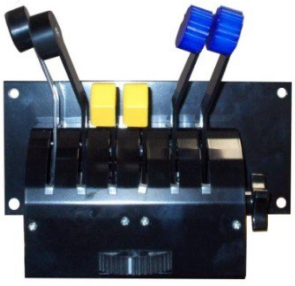


Figure 3.8: Twin Engine Throttle Quadrant – Type 4



Figure 6.1: Optional CAPS for Cirrus Configurations



Figure 6.2: Optional CAPS for Cirrus Configurations

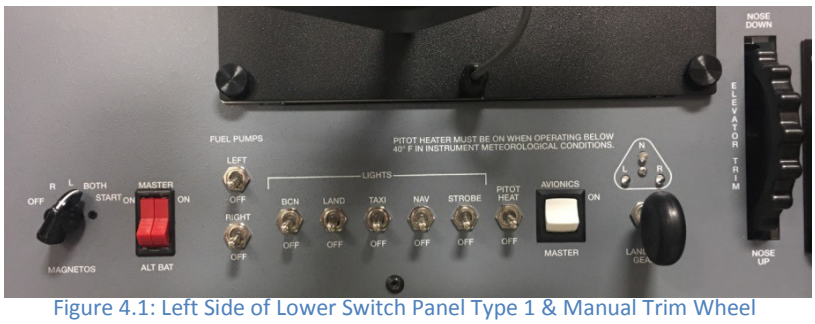


Figure 4.1: Left Side of Lower Switch Panel Type 1 & Manual Trim Wheel



Figure 4.7: Switch Panel 1 Starter Switch*



Figure 4.8: Switch Panel 1 Starter Switch*



Figure 4.2: Right Side of Lower Switch Panel Type 1*



Figure 4.9: Right Side of Lower Switch Panel Type 1*

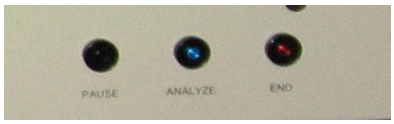


Figure 4.12: Training Session Buttons*



Figure 4.13: Training Session Buttons*



Figure 4.14: Training Session Buttons*



Figure 4.14: COM Volume/Squelch Adjustments*



Figure 4.15: COM Volume/Squelch Adjustments*



Figure 4.4: Left Side of Lower Switch Panel Type 2



Figure 4.6: Right Side of Lower Switch Panel Type 2

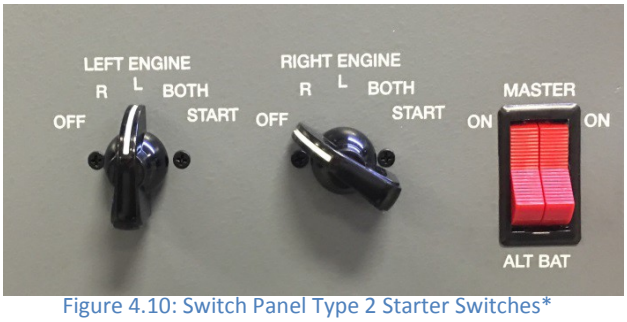


Figure 4.10: Switch Panel Type 2 Starter Switches*

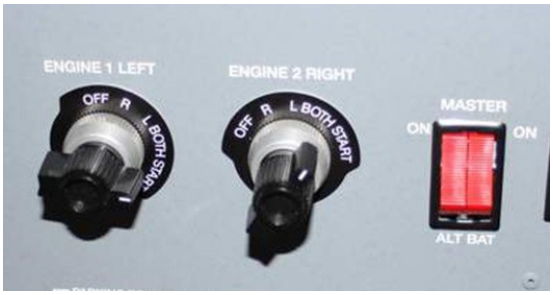


Figure 4.11: Switch Panel Type 2 Starter Switches*

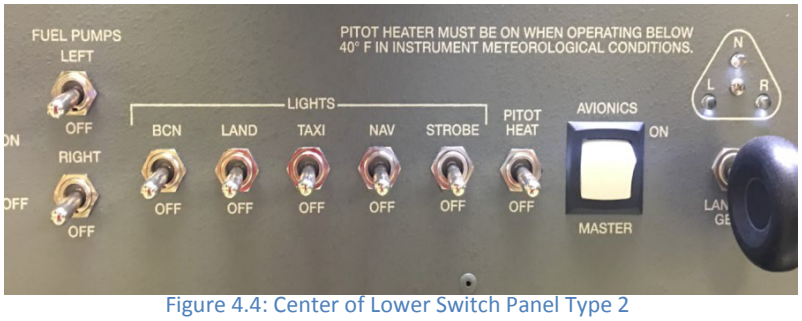


Figure 4.4: Center of Lower Switch Panel Type 2



Figure 4.5 Manual Trim Wheel



Figure 4.16: Switch Panel 1 Fuel Selector

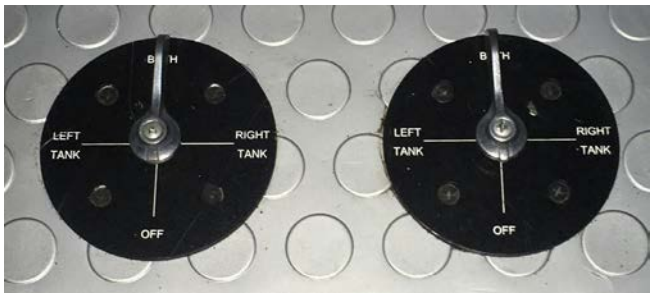


Figure 4.17: Switch Panel 2 Fuel Selector

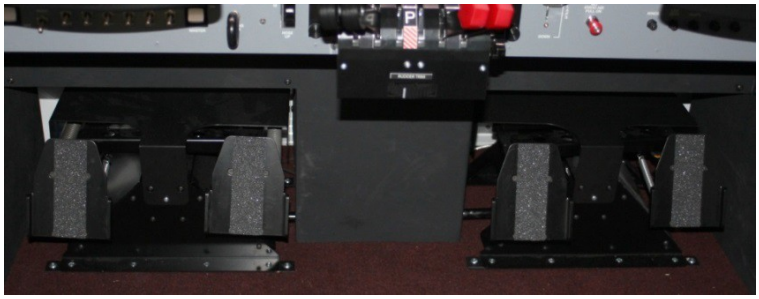


Figure 5.11: Rudder Pedals – Dual

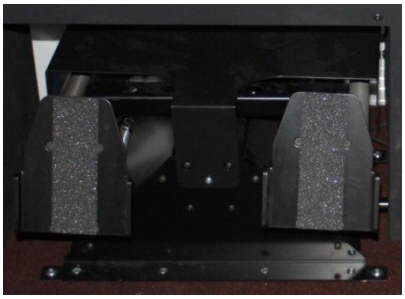


Figure 5.3: Rudder Pedals – Single

*May vary by serial number

SECTION 3: TRAINING DEVICE COMPONENTS LIST

Qty	Type	Manufacturer	Name	Description/Function	Configuration
1	Software	Microsoft	Windows XP, 7, or 10	Operating system. (depending on serial number)	All
1	Software	Microsoft	ESP or FSX	Simulation engine. (depending on serial number)	All
1	Software	Lockheed Martin	Prepar3D	Simulation engine. (depending on serial number)	All
1	Software	Redbird Flight Simulations	RB Sim	Simulation control and component integration. (depending on serial number)	All
1	Software	Redbird Flight Simulations	FMX Control	Motion system control and component integration.	FMX, MCX
1	Software	Redbird Flight Simulations	Instructors Station	Environmental, location and failure controls with map, track and glideslope display. (depending on serial number)	All
1	Software	Redbird Flight Simulations	Navigator	Simulation control and component integration. Environmental, location and failure controls with map, track and glideslope display. (depending on serial number)	All
1	Software	Redbird Flight Simulations	Parrot	Optional software providing artificial ATC services	All
1	Software	Redbird Flight Simulations	Cygnus (including Pro)	Optional software providing location services	All
1	Software	Mindstar Aviation	Redbird 1000	Virtual replication of the Garmin G1000 flight instruments, GPS, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird 430/ 530	Virtual replication of the Garmin GNS430 and GNS530	See Configuration
1	Software	Mindstar Aviation	Redbird KLN94	Virtual replication of the BendixKing KLN94	See Configuration
1	Software	Mindstar Aviation	Redbird EFD1000	Virtual replication of the Aspen EFD1000	See Configuration
1	Software	Mindstar Aviation	Redbird Entegra	Virtual replication of the Avidyne Entegra flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird Autopilot	Virtual replication of the BendixKing KAP140, Genesys STEC55x, and Garmin GFC700 (Redbird 140, 55, or 700)	See Configuration
1	Software	Mindstar Aviation	Redbird Radios	Virtual Radios replication of the BendixKing KX55 and Garmin SL40 (KX155, SL40)	See Configuration
1	Software	Mindstar Aviation	Miscellaneous Gauges	Virtual Miscellaneous Gauges	See Configuration
1	Software	Flight 1	Compass	Virtual Generic Compass	All
1	Software	Redbird Flight Simulations	Analog Gauges	Virtual Airspeed, Attitude, Altimeter, Turn and Bank, Heading, HSI, VSI, RMI, CDI, and ADF Gauges , Radios	See Configuration
1	Software	Redbird Flight Simulations	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Hardware	Redbird Flight Simulations	Simulation Computer	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
1	Hardware	Redbird Flight Simulations	Motion Control Computer	Host computer for FMX Control.	FMX, MCX
1	Hardware	Industry Standard	Instructors Station PC or Wireless Mobile Device	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Motion Platform: Type 1	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	FMX and MCX configuration - See Image 5,6

1	Hardware	Redbird Flight Simulations	Cockpit Enclosure	Cockpit enclosure to exclude distractions.	All- See Image 1,2,6,6
6	Hardware	22" LCD Industry Standard Monitor	Visual Display - Type 1	Flat Panel displays for exterior views.	All – Varies by configuration
8	Hardware	22" LCD Industry Standard Monitor	Visual Display - Type 2	Optional Flat Panel displays for exterior views.	All – Varies by configuration
6	Hardware	27" LCD Industry Standard Monitor	Visual Display - Type 3	Optional large Flat Panel displays for exterior views.	All – Varies by configuration
8	Hardware	27" LCD Industry Standard Monitor	Visual Display - Type 4	Optional large Flat Panel displays for exterior views.	All – Varies by configuration
2	Hardware	19" LCD Industry Standard Monitor	LCD	Flat Panel displays for virtual instruments.	All
1	Hardware	Redbird Flight Simulations	Yoke, Side Yoke, Center Stick (Single)	Pilot yoke with pitch and roll controller with switches and buttons for airplane systems operation. As required for each configuration.	FMX, SD, LD – Varies by configuration
1	Hardware	Redbird Flight Simulations	Yoke, Side Yoke, Center Stick (Single) – Control Loading	Optional Control loading pilot yoke with pitch and roll controller and switches and buttons for airplane systems operation. As required for each configuration.	FMX, SD, LD – Varies by configuration
1	Hardware	Redbird Flight Simulations	Yoke (Dual) – Control Loading	Control loading pilot and co-pilot yoke with pitch and roll controller with switches and buttons for airplane systems operation. As required for each configuration.	MCX / MCX-SD
1	Hardware	Redbird Flight Simulations	Throttle Quadrant	As required for each configuration.	All – See Configuration
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Single)	Pilot rudder control pedals with toe brakes.	FMX, SD, LD – Varies by configuration
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Single) – Control Loading	Optional Control loading pilot rudder control pedals with toe brakes.	FMX, SD, LD – Varies by configuration
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Dual)	Pilot and co-pilot rudder control pedals with toe brakes.	MCX / MCX-SD – Varies by configuration
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Dual) – Control Loading	Optional Control loading Pilot and co-pilot rudder control pedals with toe brakes.	MCX / MCX-SD – Varies by configuration
1	Hardware	Redbird Flight Simulations	Switch Panel: Type 1	Lower switch panel with airplane configuration and systems controls.	FMX, SD, LD
1	Hardware	Redbird Flight Simulations	Switch Panel: Type 2	Lower switch panel with airplane configuration and systems controls.	MCX / MCX-SD
1	Hardware	Redbird Flight Simulations	Instrument Controls Overlay	Flight instruments, radios, airplane configuration and systems controls as required for each configuration.	All – See Configuration
1	Hardware	Redbird Flight Simulations	CAPS	Optional CAPS	All – See Configuration

Table 1: Training Device Component List (Above)

Statements of Compatibility of Software and Hardware

AC 61-136B, Appendix A.6.2

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows (Operating System)

“This is to certify that Microsoft Corporation, the owner and developer of the Windows operating system, has evaluated that their operating system works with industry standard PC’s and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.”

Lockheed Martin Prepar3D (Simulation Engine)

This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC’s and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software

Mindstar Aviation Redbird Virtual Instrument Suite

This is to certify that Mindstar Aviation, the owner and developer of the Redbird Virtual Instrument Suite, including the Redbird 1000, Redbird 430/530, Redbird KLN94, Redbird 140 Autopilot, Redbird 55 Autopilot, Redbird Radios, Redbird Entegra and Redbird Perspective and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Redbird Virtual Instrument Suite software package is fully compatible with the Redbird Flight Simulations LD, SD, FMX, MCX and all configurations of those models. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations**.**

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Flight 1 Compass

"This is to certify that Flight 1, the owner and developer of the Compass and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Compass software is fully compatible with the Redbird Flight Simulations LD, SD, FMX, MCX and all configurations of those models. Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

SECTION 4: AVIATION TRAINING DEVICE (ATD) DESIGN CRITERIA LIST

The following section provides the detailed "word for word" listing and design criteria of each of the required items, functions, and capabilities (listed in AC 61-136B, for BATD requirements Appendix B and the additional AATD items of Appendix C) and operational performance value/scale (as applicable) for each of the functions described for the Redbird LD, SD, FMX, MCX/SD and MCX models.

Basic ATD Requirements List [Appendix B items]

All configurations for this model, as noted, meet AC 61-136B, Appendix B requirements

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following (General Control Requirements):

B.3.1.1 The aircraft physical flight and associated control systems must be recognizable as to their function and how they are to be manipulated solely from their appearance. These physical flight control systems cannot use interfaces such as a keyboard, mouse, or gaming joystick to control the aircraft in simulated flight.

B.3.1.2 Virtual controls are those controls used to set up certain aspects of the simulation (such as selecting the aircraft configuration, location, weather conditions, etc.) and otherwise program, effect, or pause the training device. These controls are often part of the instructor station or independent computer interface.

B.3.1.3 Except for the initial setup, a keyboard or mouse may not be used to set or position any feature of the ATD flight controls for the maneuvers or training tasks to be accomplished. See the control requirements listed below as applicable to the aircraft model represented. The pilot must be able to operate the controls in the same manner as it would be in the actual aircraft. This includes the landing gear, wing flaps, cowl flaps, carburetor heat, mixture, propeller, and throttle controls appropriate to the aircraft model represented.

B.3.1.4 The physical arrangement, appearance, and operation of controls, instruments, and switches required by this appendix should closely model the aircraft represented. Manufacturers are expected to recreate the appearance, arrangement, operation, and function of realistically placed physical switches and other required controls representative of an aircraft instrument panel that includes at least the following:

- Master/battery;
- Magnetos for each engine (as applicable);
- Alternators or generators for each engine;
- Auxiliary power unit (APU) (if applicable);
- Fuel boost pumps/prime boost pumps for each engine;
- Avionics master;
- Pitot heat; and
- Rotating beacon/strobe, navigation, taxi, and landing lights. Only the software evaluated and approved by the FAA is installed on this computer system.

B.3.1.5 When an FAA-approved ATD is in use, only the software evaluated by the FAA may be loaded for use on that computer system. This does not preclude providing software updates that do not otherwise change the appearance of the systems operation.

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following Control Requirements (For Airplane):

B.3.2.1.1 A self-centering displacement yoke or control stick that allows continuous adjustment of pitch and bank.

B.3.2.1.2 Self-centering rudder pedals that allow continuous adjustment of yaw and corresponding reaction in heading and roll.

B.3.2.1.3 Throttle or power control(s) that allows continuous movement from idle to full-power settings and corresponding changes in pitch and yaw, as applicable.

B.3.2.1.4 Mixture/condition, propeller, and throttle/power control(s) as applicable to the M/M of aircraft represented.

B.3.2.1.5 Controls for the following items, as applicable to the category and class of aircraft represented:

- Wing flaps,
- Pitch trim,

- Communication and navigation radios,
- Clock or timer,
- Gear handle (if applicable),
- Transponder,
- Altimeter,
- Carburetor heat (if applicable), and
- Cowl flaps (if applicable).

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following (Control Input Functionality and Response Criteria):

B.3.3.1 Time from control input to recognizable system response must be without delay (i.e., not appear to lag in any way). “Redbird Flight Simulations, Inc.” verifies that the “Redbird LD, SD, FMX, and MCX (MCX/SD)” meets this requirement.

B.3.3.2 The control inputs must be tested by the computer and software program at each startup and displayed as a confirmation message of normal operation or a warning message that the transport delay time or any design parameter is out of tolerance. It should not be possible to continue the training session unless the problem is resolved and all components are functioning properly.

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following (Display Requirements):

B.3.4.1 The following instruments and indicators must be replicated and properly located as appropriate to the aircraft represented:

B.3.4.1.1 Flight instruments in a standard configuration representing the traditional “round” dial flight instruments. An electronic primary flight display (PFD) with reversionary and backup flight instruments is also acceptable.

B.3.4.1.2 A sensitive altimeter with incremental markings each 20 feet or less, operable throughout the normal operating range of the M/M of aircraft represented.

B.3.4.1.3 A magnetic direction indicator.

B.3.4.1.4 A heading indicator with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the M/M of aircraft represented.

B.3.4.1.5 An airspeed indicator with incremental markings as shown for the M/M aircraft represented; airspeed markings of less than 20 knots need not be displayed.

B.3.4.1.6 A vertical speed indicator (VSI) with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ±2,000 fpm total display, or as applicable to the M/M of aircraft being represented.

B.3.4.1.7 A gyroscopic rate-of-turn indicator or equivalent with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.

B.3.4.1.8 A slip and skid indicator with coordination information displayed in the conventional inclinometer format where a coordinated flight condition is indicated with the ball in the center position. A split image triangle indication as appropriate for PFD configurations may be used.

B.3.4.1.9 An attitude indicator with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to M/M of aircraft represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.

B.3.4.1.10 Engine instruments as applicable to the M/M of aircraft being represented, providing markings for the normal ranges including the minimum and maximum limits.

B.3.4.1.11 A suction gauge or instrument pressure gauge with a display applicable to the aircraft represented.

B.3.4.1.12 A flap setting indicator that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.

B.3.4.1.13 A pitch trim indicator with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.

B.3.4.1.14 Communication radio(s) with a full range of selectable frequencies displaying the radio frequency in use.

B.3.4.1.15 Navigation radio(s) with a full range of selectable frequencies displaying the frequency in use and capable of replicating both precision and nonprecision instruments, including approach procedures (each with an aural identification feature), and a marker beacon receiver. For example, an instrument landing system (ILS), non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or very high frequency omni-directional range (VOR). Graduated markings as indicated below must be present on each course deviation indicator (CDI) as applicable. The marking should include:

- One-half dot or less for course/glideslope (GS) deviation (i.e., VOR, LOC, or ILS), and
- Five degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable.

B.3.4.1.16 A clock with incremental markings for each minute and second, or a timer with a display of minutes and seconds.

B.3.4.1.17 A transponder that displays the current transponder code.

B.3.4.1.18 A fuel quantity indicator(s) that displays the fuel remaining, either in analog or digital format, appropriate for M/M of aircraft represented.

B.3.4.2 All instrument displays listed above must be visible during all flight operations. Allowances can be made for multifunction electronic displays that may not display all instruments simultaneously. All of the displays must provide an image of the instrument that is clear and:

B.3.4.2.1 Does not appear to be out of focus or illegible.

B.3.4.2.2 Does not appear to "jump" or "step" during operation.

B.3.4.2.3 Does not appear with distracting jagged lines or edges.

B.3.4.2.4 Does not appear to lag relative to the action and use of the flight controls.

B.3.4.3 Control inputs should be reflected by the flight instruments in real time and without a perceived delay in action. Display updates must show all changes (within the total range of the replicated instrument) that are equal to or greater than the values stated below:

B.3.4.3.1 Airspeed indicator: change of 5 knots.

B.3.4.3.2 Attitude indicator: change of 2 degrees in pitch and bank.

B.3.4.3.3 Altimeter: change of 10 feet.

B.3.4.3.4 Turn and bank: change of ¼ standard rate turn.

B.3.4.3.5 Heading indicator: change of 2 degrees.

B.3.4.3.6 VSI: change of 100 fpm.

B.3.4.3.7 Tachometer: change of 25 rpm or 2 percent of turbine speed.

B.3.4.3.8 VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS.

B.3.4.3.9 ADF: change of 2 degrees.

B.3.4.3.10 GPS: change as appropriate for the model of GPS-based navigator represented.

B.3.4.3.11 Clock or timer: change of 1 second.

B.3.4.4 Displays must reflect the dynamic behavior of an actual aircraft (e.g., a VSI reading of 500 fpm must reflect a corresponding movement in altitude, and an increase in power must reflect an increase in the rpm indication or power indicator.)

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following (Flight Dynamics Requirements):

B.3.5.1 Flight dynamics of the ATD should be comparable to the way the represented training aircraft performs and handles. However, there is no requirement for an ATD to have control loading to exactly replicate any particular aircraft.

B.3.5.2 Aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate, and hovering/sideward/forward/rearward flight) should be comparable to the aircraft being represented. A performance table will need to be included in the QAG for each aircraft configuration for sea level and 5,000 feet using standard atmosphere and gross weight conditions. An alternate performance altitude for 6,000 feet can be used if the manufacturer of that aircraft has a performance chart reflecting that altitude; otherwise the

ATD manufacturer will need to interpolate the performance for the chart. Performance at altitude for turboprop or turbojet configurations should reflect 18,000 ft.

B.3.5.3 Aircraft vertical lift component must change as a function of bank comparable to the way the aircraft being represented performs and handles.

B.3.5.4 Changes in flap setting, slat setting, gear position, collective control, or cyclic control must be accompanied by changes in flight dynamics comparable to the way the M/M of aircraft represented performs and handles.

B.3.5.5 The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft represented performs and handles.

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following (Instructional Management Requirements):

B.3.6.1 The instructor must be able to pause the system at any time during the training simulation for the purpose of administering instruction or procedural recommendations.

B.3.6.2 If a training session begins with the “aircraft in the air” and ready for the performance of a particular procedural task, the instructor must be able to manipulate the following system parameters independently of the simulation:

- Aircraft geographic location,
- Aircraft heading,
- Aircraft airspeed,
- Aircraft altitude, and
- Wind direction, speed, and turbulence.

B.3.6.3 The system must be capable of recording both a horizontal and vertical track of aircraft movement during the entire training session for later playback and review.

B.3.6.4 The instructor must be able to disable any of the instruments prior to or during a training session and be able to simulate failure of any of the instruments without stopping or freezing the simulation to affect the failure. This includes simulated engine failures and the following aircraft systems failures: alternator or generator, vacuum or pressure pump, pitot static, electronic flight displays, or landing gear or flaps, as appropriate.

B.3.6.5 The ATD must have at least a navigational area database that is local (25 nautical miles (NM)) to the training facility to allow reinforcement of procedures learned during actual flight in that area. All navigational data must be based on procedures as published per 14 CFR part 97.

Advanced ATD Requirements List [Appendix C items]

All configurations, as noted in AC 61-136B, Appendix C meet these additional AATD design criteria items listed.

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following additional AATD CRITERIA:

C.3.1.1 A realistic shrouded (enclosed) or unshrouded (open) cockpit design and instrument panel arrangement representing a specific model aircraft cockpit.

C.3.1.2 Cockpit knobs, system controls, switches, and/or switch panels in realistic sizes and design appropriate to each intended functions, in the proper position and distance from the pilot's seated position, and representative of the category and class of aircraft being represented.

C.3.1.3 Primary flight and navigation instruments appropriately sized and properly arranged that exhibit neither stepping nor excessive transport delay.

C.3.1.4 Digital Avionics Panel

C.3.1.5 Global Positioning System (GPS) navigator with moving map display.

C.3.1.6 Two-axis autopilot, and, as appropriate, a flight director (FD). This is only required when an autopilot is original standard equipment from the aircraft manufacturer.

C.3.1.7 Pitch trim (manual or electric pitch trim) permitting indicator movement either electrically or analog in an acceptable trim ratio (airplane only).

C.3.1.8 An independent visual system, panel, or screen that provides realistic cues in both day and night visual flight rules (VFR) and instrument flight rules (IFR) meteorological conditions to enhance a pilot's visual orientation in the vicinity of an airport including:

- Adjustable visibility parameters; and
- Adjustable ceiling parameters.

C.3.1.9 A fixed pilot seat appropriate to the aircraft configuration, including an adjustable height and an adjustable forward and aft seat position.

C.3.1.10 Rudder pedals secured to the cockpit floor structure, or that can be physically secured to the floor beneath the device in proper relation to cockpit orientation.

C.3.1.11 Push-to-talk switch on the control yoke.

C.3.1.12 A separate instructor station to permit effective interaction without interrupting the flight in overseeing the pilot's horizontal and vertical flight profiles in real time and space. This must include the ability to:

1. Oversee tracks along airways, holding entries and patterns, and Localizer (LOC) and glideslope (GS) alignment/deviation (or other approaches with a horizontal and vertical track).
2. Function as air traffic control (ATC) in providing vectors, etc., change in weather conditions, ceilings, visibilities, wind speed and direction, light/moderate/severe turbulence, and icing conditions.
3. Invoke failures in navigation and instruments, radio receivers, landing gear and flaps, engine power (partial and total), and other aircraft systems (pitot, electric, static, etc.) by using either a keyboard or mouse.

The Redbird LD, SD, FMX, MCX/SD and MCX models meet the following additional encouraged (not required) AATD CRITERIA:

C.3.2.1 Multi-panel or wrap-around visual system providing a 120 degrees or more of horizontal vision.

C.3.2.2 Automated ATC communications, scenario-based training (SBT), or line-oriented type training in which the instructor can evaluate pilot performance without having to act as ATC.

C.3.2.3 Simulated loss of performance and aerodynamic changes from ice accretion.

C.3.2.4 Realistic aircraft engine sound appropriate to the aircraft configuration, power settings, and speed.

C.3.2.5 A magnetic compass with incremental markings each 5 degrees, that displays the proper lead or lag during turns, and displays incremental markings typical of that shown in the aircraft.

SECTION 5: AIRCRAFT CONFIGURATIONS

List of Previously Approved Configurations (QAG v4.5E):

Beechcraft BE36-S1



MTOW	FWD/AFT CG Limit
3600 lbs	81 in / 87.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Beechcraft BE36-G1



MTOW	FWD/AFT CG Limit
3600 lbs	81 in / 87.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Beechcraft BE55-S1



MTOW	FWD/AFT CG Limit
5000 lbs	80 in / 86 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (PP-TT-MM) (See Figure 3.6)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Beechcraft BE58-S1



MTOW	FWD/AFT CG Limit
5500 lbs	78 in / 86 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (PP-TT-MM) (See Figure 3.6)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Beechcraft BE58-S2



MTOW	FWD/AFT CG Limit
5500 lbs	78 in / 86 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Beechcraft BE58-G1



MTOW	FWD/AFT CG Limit
5500 lbs	78 in / 86 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Beechcraft BE76-S1



MTOW	FWD/AFT CG Limit
3900 lbs	110.6 in / 117.5 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Cessna C172-S1



MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 3.1)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Cessna C172-S2



MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 3.1)**
- **Traditional analog gauges & KLN94 GPS with KAP140 Autopilot**

Cessna C172-G1



MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 3.1)**
- **Glass Cockpit G1000 PFD/MFD & KAP140 Autopilot**

Cessna C172-G2



MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 3.1)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Cessna C182-S1



MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Cessna C182-S2



MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Traditional analog gauges & KLN94 GPS with KAP140 Autopilot**

Cessna C182-G1



MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Glass Cockpit G1000 PFD/MFD & KAP140 Autopilot**

Cessna C182-G2



MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Cessna C206-S1



MTOW	FWD/AFT CG Limit
3600 lbs	42.5 in / 49.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Cessna C206-S2



MTOW	FWD/AFT CG Limit
3600 lbs	42.5 in / 49.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Traditional analog gauges & KLN94 GPS with KAP140 Autopilot**

Cessna C206-G1



MTOW	FWD/AFT CG Limit
3600 lbs	42.5 in / 49.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Glass Cockpit G1000 PFD/MFD & KAP140 Autopilot**

Cessna C206-G2



MTOW	FWD/AFT CG Limit
3600 lbs	42.5 in / 49.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Cessna C206T-S1



MTOW	FWD/AFT CG Limit
3600 lbs	42.5 in / 49.7 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Cirrus SR20-G1



MTOW	FWD/AFT CG Limit
3000 lbs	144.1 in / 148 in.

- **Yoke – Left (See Figure 2.2)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Glass Cockpit Entegra PFD/MFD & Dual GNS430 GPS with STEC55 Autopilot**

Cirrus SR20-G2



MTOW	FWD/AFT CG Limit
3000 lbs	144.1 in / 148 in.

- **Yoke – Left (See Figure 2.2)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Glass Cockpit Perspective PFD/MFD & FMS Keypad with Perspective Autopilot**

Cirrus SR22-G1



MTOW	FWD/AFT CG Limit
3400 lbs	142.3 in / 148.1 in

- **Yoke – Left (See Figure 2.2)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Glass Cockpit Entegra PFD/MFD & Dual GNS430 GPS with STEC55 Autopilot**

Cirrus SR22-G2



MTOW	FWD/AFT CG Limit
3400 lbs	142.3 in / 148.1 in

- **Yoke – Left (See Figure 2.2)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Glass Cockpit Perspective PFD/MFD & FMS Keypad with Perspective Autopilot**

Diamond DA20-S1



MTOW	FWD/AFT CG Limit
1609 lbs	9.84 in / 15.4 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & GNS430 GPS with STEC55 Autopilot**

Diamond DA40-G1



MTOW	FWD/AFT CG Limit
2535 lbs	96.9 in / 102.0 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Diamond DA40-G2



MTOW	FWD/AFT CG Limit
2535 lbs	96.9 in / 102.0 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit G1000 PFD/MFD & KAP140 Autopilot**

Diamond DA42-G1



MTOW	FWD/AFT CG Limit
3935 lbs	95.47 in / 96.5 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Twin Engine Lever (TT-MM) (See Figure 3.7)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Diamond DA42L-G2



MTOW	FWD/AFT CG Limit
3935 lbs	95.47 in / 96.5 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Twin Engine Lever (TT-PP-MM) (See Figure 3.5)**
- **Glass Cockpit G1000 PFD/MFD & KAP140 Autopilot**

Liberty LXL2-S1



MTOW	FWD/AFT CG Limit
1653 lbs	82.2 in / 86.8 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & GNS430/530 GPS with STEC55 Autopilot**

Liberty LXL2-S2



MTOW	FWD/AFT CG Limit
1653 lbs	82.2 in / 86.8 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & GNS430 GPS, EFD1000, SL40 & STEC55 Autopilot**

Mooney M20-G1



MTOW	FWD/AFT CG Limit
3668 lbs	46 in / 51 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Piper Warrior PA28-60-S1



MTOW	FWD/AFT CG Limit
2440 lbs	88.3 in / 93.0 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & Dual GNS430 GPS with STEC55 Autopilot**

Piper Warrior PA28-60-S2



MTOW	FWD/AFT CG Limit
2440 lbs	88.3 in / 93.0 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Piper Warrior PA28-60-G1



MTOW	FWD/AFT CG Limit
2440 lbs	88.3 in / 93.0 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Glass Cockpit Entegra PFD/MFD & Dual GNS430 GPS with STEC55 Autopilot**

Piper Archer PA28-80-S1



MTOW	FWD/AFT CG Limit
2550 lbs	88.6 in / 93.0 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & Dual GNS430 GPS with STEC55 Autopilot**

Piper Archer PA28-80-S2



MTOW	FWD/AFT CG Limit
2550 lbs	88.6 in / 93.0 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Lever (T-M) (See Figure 3.3)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Piper Archer PA28-80-G1



MTOW	FWD/AFT CG Limit
2550 lbs	88.6 in / 93.0 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Piper Arrow PA28-2R-S1



MTOW	FWD/AFT CG Limit
2750 lbs	88.9 in / 91.5 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Traditional analog gauges & Dual GNS430 GPS with STEC55 Autopilot**

Piper Arrow PA28-2R-S2



MTOW	FWD/AFT CG Limit
2750 lbs	88.9 in / 91.5 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Piper Arrow PA28-2R-G1



MTOW	FWD/AFT CG Limit
2750 lbs	88.9 in / 91.5 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit Entegra PFD/MFD & Dual GNS430 GPS with STEC55 Autopilot**

Piper Arrow PA28-2R-G2



MTOW	FWD/AFT CG Limit
2750 lbs	88.9 in / 91.5 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.4)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Piper Seneca PA34-S1



MTOW	FWD/AFT CG Limit
4750 lbs	90.6 in / 94.6 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Piper Seneca PA34-G1



MTOW	FWD/AFT CG Limit
4750 lbs	90.6 in / 94.6 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Piper Seminole PA44-S1



MTOW	FWD/AFT CG Limit
3800 lbs	89 in / 93 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Piper Seminole PA44-G1



MTOW	FWD/AFT CG Limit
3800 lbs	89 in / 93 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Piper Seminole PA44T-S1



MTOW	FWD/AFT CG Limit
3925 lbs	89.7 in / 93 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Socata TB10-S1



MTOW	FWD/AFT CG Limit
2535 lbs	42.6 in / 47.4 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Single Engine Complex Vernier (T-P-M) (See Figure 3.2)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Tecnam P2006-G1



MTOW	FWD/AFT CG Limit
2601 lbs	8.7 in / 16.3 in

- **Yoke – Stick (See Figure 2.3)**
- **Throttle – Twin Engine Lever (TT-CC-PP) (See Figure 3.8)**
- **Glass Cockpit G1000 PFD/MFD & STEC55 Autopilot**

Additional configurations included in this version (QAG v4.6D)

Piper Seneca PA34T-S1



MTOW	FWD/AFT CG Limit
4407 lbs	88.0 in / 94.6 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Traditional analog gauges & GNS430/530 GPS with KAP140 Autopilot**

Piper Seneca PA34T-G1



MTOW	FWD/AFT CG Limit
4407 lbs	88.0 in / 94.6 in

- **Yoke – Center (See Figure 2.1)**
- **Throttle – Twin Engine Complex Lever (TT-PP-MM) (See Figure 3.5)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

Performance Table

Aircraft Model	V_{SO}	V_{S1}	V_X	V_Y	V_A	V_{NE}	V_{MCA}	KTAS @ Cruise / 75% power setting	Rate of Climb (fpm) @ (V_Y) / Full Power	Single Engine Rate of Climb (V_{YSE})
Beechcraft (A36) Bonanza	53 KIAS	62 KIAS	78 KIAS	96 KIAS	140 KIAS	204 KIAS	N/A	160 KTAS	1030 fpm	N/A
							5000' -->	166 KTAS	740 fpm	N/A
Cessna 172 Skyhawk	40 KIAS	48 KIAS	56 KIAS	74 KIAS	105 KIAS	163 KIAS	N/A	116 KTAS	730 fpm	N/A
							5000' -->	119 KTAS	550 fpm	N/A
Cessna 182 Skylane	35 KIAS	40 KIAS	58 KIAS	80 KIAS	110 KIAS	175 KIAS	N/A	129 KTAS	925 fpm	N/A
							5000' -->	135 KTAS	705 fpm	N/A
Cessna 206 Stationair	34 KIAS	41 KIAS	65 KIAS	84 KIAS	120 KIAS	183 KIAS	N/A	141 KTAS	920 fpm	N/A
							5000' -->	145 KTAS	640 fpm	N/A
Cessna 206 Stationair TC	39 KIAS	50 KIAS	74 KIAS	87 KIAS	125 KIAS	182 KIAS	N/A	139 KTAS	1050 fpm	N/A
							5000' -->	144 KTAS	955 fpm	N/A
Cirrus SR20	56 KIAS	65 KIAS	75 KIAS	85 KIAS	131 KIAS	200 KIAS	N/A	147 KTAS	828 fpm	N/A
							5000' -->	151 KTAS	566 fpm	N/A
Cirrus SR22	59 KIAS	70 KIAS	78 KIAS	91 KIAS	133 KIAS	201 KIAS	N/A	170 KTAS	1304 fpm	N/A
							5000' -->	175 KTAS	1015 fpm	N/A
Diamond DA20	37 KIAS	41 KIAS	57 KIAS	65 KIAS	104 KIAS	161 KIAS	N/A	98 KTAS	675 fpm	N/A
							5000' -->	108 KTAS	440 fpm	N/A
Diamond DA40	49 KIAS	52 KIAS	66 KIAS	66 KIAS	108 KIAS	178 KIAS	N/A	128 KTAS	1100 fpm	N/A
							5000' -->	136 KTAS	720 fpm	N/A
Liberty XL2	41 KIAS	51 KIAS	70 KIAS	80 KIAS	100 KIAS	157 KIAS	N/A	N/A	682 fpm	N/A
							5000' -->	N/A	438 fpm	N/A
Mooney M20	61 KIAS	67 KIAS	84 KIAS	104 KIAS	127 KIAS	194 KIAS	N/A	182 KTAS	1375 fpm	N/A
							5000' -->	197 KTAS	1275 fpm	N/A
Piper PA-28-161 Warrior III	44 KIAS	50 KIAS	53 KIAS	79 KIAS	111 KIAS	160 KIAS	N/A	106 KTAS	644 fpm	N/A
							5000' -->	114 KTAS	396 fpm	N/A
Piper PA-28-181 Archer III	45 KIAS	50 KIAS	60 KIAS	76 KIAS	113 KIAS	154 KIAS	N/A	121 KTAS	680 fpm	N/A
							5000' -->	125 KTAS	480 fpm	N/A
Piper PA-28R-201 Arrow	55 KIAS	60 KIAS	60 KIAS	90 KIAS	118 KIAS	183 KIAS	N/A	131 KTAS	840 fpm	N/A
							5000' -->	135 KTAS	580 fpm	N/A

Aircraft Model	V_{SO}	V_{S1}	V_X	V_Y	V_A	V_{NE}	V_{MCA}	KTAS @ Cruise / 75% power setting	Rate of Climb (fpm) @ (V_Y) / Full Power	Single Engine Rate of Climb (V_{YSE})
Socata TB10	53 KIAS	60 KIAS	65 KIAS	78 KIAS	122 KIAS	165 KIAS	N/A	119 KTAS	787 fpm	N/A
							5000' -->	124 KTAS	525 fpm	N/A
Beechcraft (BE55) Baron	73 KIAS	79 KIAS	91 KIAS	107 KIAS	157 KIAS	224 KIAS	78 KIAS	168 KTAS	1725 fpm	400 fpm @ 100 KIAS
							5000' -->	180 KTAS	1250 fpm	125 fpm @ 100 KIAS
Beechcraft (BE58) Baron	74 KIAS	84 KIAS	92 KIAS	105 KIAS	156 KIAS	223 KIAS	84 KIAS	188 KTAS	1725 fpm	395 fpm @ 101 KIAS
							5000' -->	203 KTAS	1325 fpm	150 fpm @ 101 KIAS
Beechcraft (BE76) Duchess	60 KIAS	70 KIAS	71 KIAS	85 KIAS	132 KIAS	194 KIAS	65 KIAS	148 KTAS	1250 fpm	230 fpm @ 85 KIAS
							5000' -->	160 KTAS	900 fpm	90 fpm @ 85 KIAS
Diamond DA42 L360	57 KIAS	64 KIAS	90 KIAS	90 KIAS	126 KIAS	194 KIAS	65 KIAS	153 KTAS	1119 fpm	185 fpm @ 85 KIAS
							5000' -->	160 KTAS	1045 fpm	106 fpm @ 85 KIAS
Diamond DA42 NG	62 KIAS	69 KIAS	85 KIAS	90 KIAS	122 KIAS	188 KIAS	76 KIAS	147 KTAS	1065 fpm	175 fpm @ 85 KIAS
							5000' -->	152 KTAS	996 fpm	122 fpm @ 85 KIAS
Piper PA-34 Seneca	61 KIAS	67 KIAS	76 KIAS	88 KIAS	139 KIAS	204 KIAS	66 KIAS	157 KTAS	1470 fpm	255 fpm @ 88 KIAS
							5000' -->	171 KTAS	1380 fpm	220 fpm @ 88 KIAS
Piper PA-34T Seneca	62 KIAS	64 KIAS	76 KIAS	92 KIAS	136 KIAS	205 KIAS	66 KIAS	159 KTAS	1500 fpm	290 fpm @ 92 KIAS
							5000' -->	169 KTAS	1500 fpm	260 fpm @ 92 KIAS
Piper PA-44 Seminole	55 KIAS	57 KIAS	70 KIAS	88 KIAS	135 KIAS	202 KIAS	56 KIAS	152 KTAS	1350 fpm	225 fpm @ 88 KIAS
							5000' -->	159 KTAS	925 fpm	0 fpm @ 88 KIAS
Piper PA-44T Seminole	56 KIAS	60 KIAS	84 KIAS	88 KIAS	137 KIAS	202 KIAS	57 KIAS	151 KTAS	1275 fpm	180 fpm @ 88 KIAS
							5000' -->	159 KTAS	1175 fpm	150 fpm @ 88 KIAS
Tecnam P2006	53 KIAS	66 KIAS	80 KIAS	80 KIAS	118 KIAS	167 KIAS	62 KIAS	142 KTAS	1150 fpm	230 fpm @ 80 KIAS
							5000' -->	139 KTAS	840 fpm	60 fpm @ 80 KIAS

SECTION 6: VISUAL SYSTEM WITH IFR, VFR, DAY, AND NIGHT CAPABILITY

Redbird LD, SD, FMX, MCX Visual System
The visual system is capable of providing a field-of-view of a minimum of 45 degrees horizontally and 30 degrees vertically, simultaneously for each pilot, including adjustable cloud base and visibility in night, dusk and day scenes.

- The Redbird LD, SD, FMX, MCX provides a means of recording the visual response time for the visual system that is installed.
- The Redbird LD, SD, FMX, MCX visual system is free of optical discontinuities and artifacts that create non-realistic cues.
- The visual system is directly displayed on six (6) or (8) LCD monitors inside the cockpit enclosure, situated in an arc around the Pilot. Each monitor is 28 cm tall, and 47 cm wide (36.5 cm x 63.5cm OPTIONAL). Based upon the designated Pilot Eye Point, these monitors provide a horizontal FOV of at least 220 (6 monitors) to 260 (8 monitors) degrees and a vertical FOV of minimally 30 degrees.

Daylight: The visual system provides full color presentations and sufficient surfaces with appropriate textural cues to conduct a visual approach, landing and airport movement. Surface shading effects are consistent with the simulated sun position.

Twilight: The visual system provides full color presentations of reduced ambient intensity, sufficient surfaces with appropriate textural cues that include self-illuminated objects such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement. Scenes include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by representative ownship lighting.

Night: The visual system provides the same as above except the portrayal of reduced ambient intensity; therefore, there is no ground cues that are not self-illuminating or illuminated by ownship lights.

Designated Eye Point: The designated Pilot Eye Point is located 52 cm from the center of the forward most external view monitor, 61 cm from the left most external view monitor and 24 cm from the ceiling of the simulator enclosure. This point is roughly centered over the pilot's seat when it is adjusted to the forward most position, at a height consistent with the height of the pilot's head.

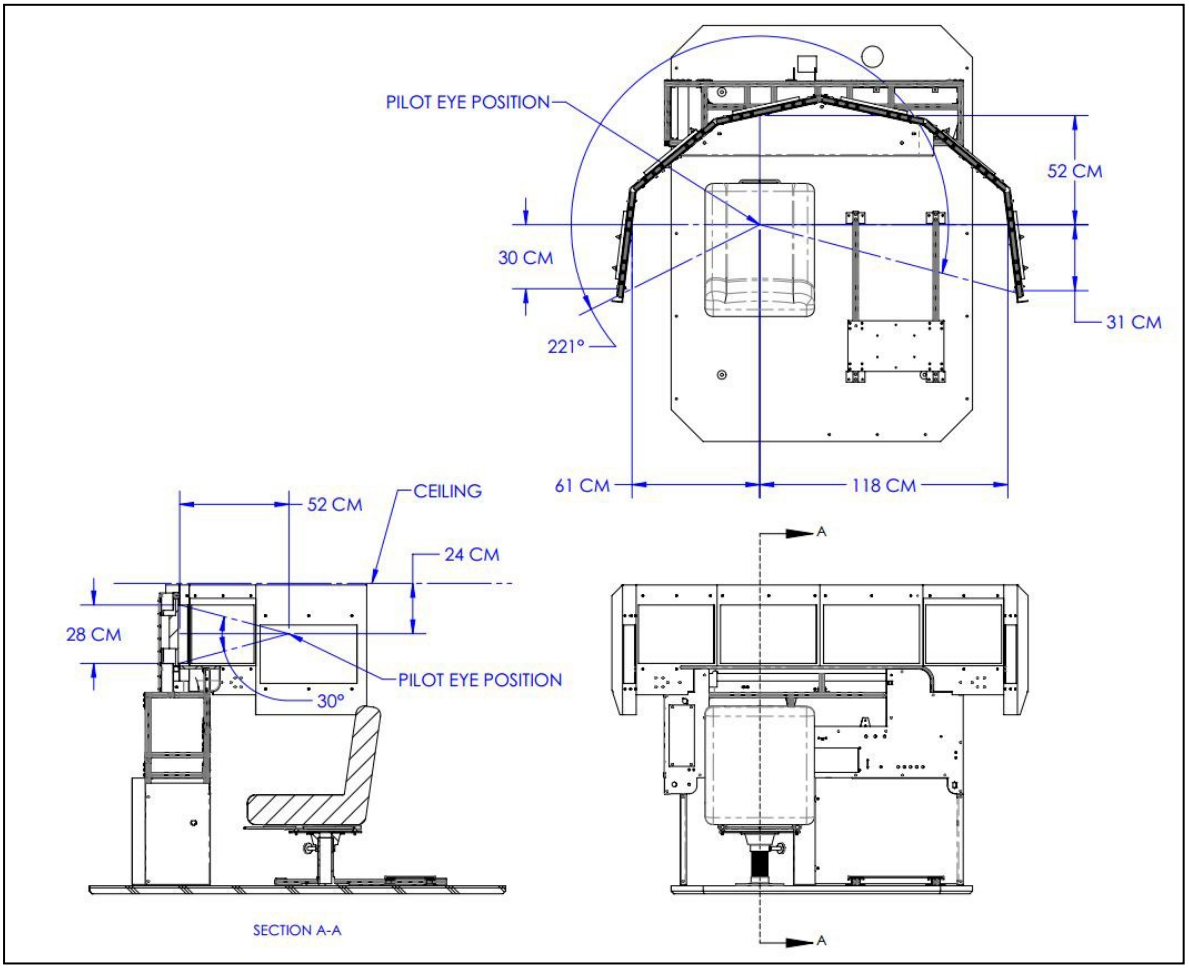


Figure 1: Designated Eye Point Diagram

Figure 5.4: Visual Displays – Type 1 (22” Display Monitors)



Figure 5.5: Visual Displays – Type 2 (22” Display Monitors)

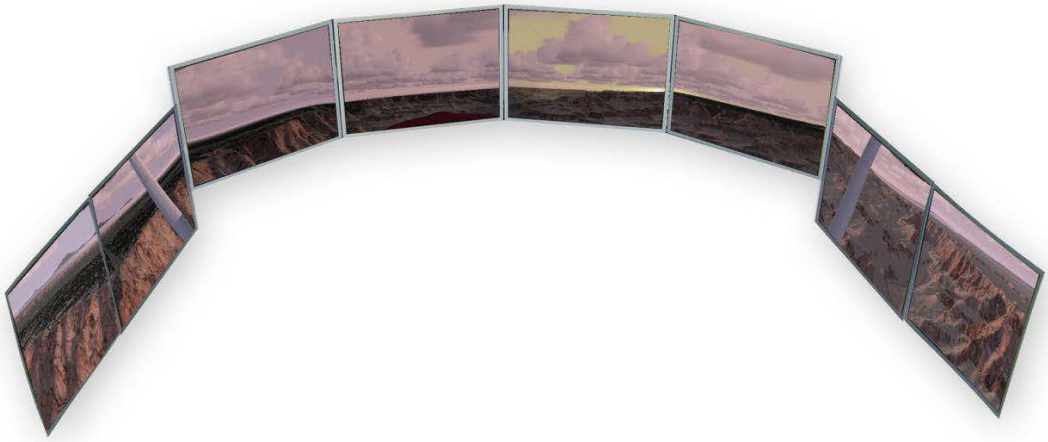


Figure 5.8: Visual Displays – Type 3 (27” Display Monitors)



Figure 5.9: Visual Displays – Type 4 (27” Display Monitors)



Figure 5.6: Example Assembled Cockpit Configuration



Figure 5.7: Example Assembled Cockpit Configuration



SECTION 7: ATD FUNCTIONS & MANEUVERS CHECKLISTS

PROCEDURES AND TASKS TEST CHECKLIST

Maneuvers and Tasks	Yes/No/NA
a) Pretakeoff	
1) Engine start	Yes
2) Taxi and brake operation	Yes
b) Takeoff	
1) AIRPLANE Takeoff	
i) Run-up and powerplant checks	Yes
ii) Acceleration characteristics	Yes
iii) Nosewheel and rudder steering	Yes
iv) Effect of crosswind	Yes
v) Instrument	Yes
vi) Landing gear, wing flap operation	Yes
2) HELICOPTER Takeoff	
i) Powerplant checks	NA
ii) From hover	NA
iii) From ground	NA
Maneuvers and Tasks	**Yes/No/NA**
iv) Vertical	NA
v) Running	NA
c) In-Flight Operation	
1) AIRPLANE In-Flight Operation	
i) Climb	Yes
(a) Normal and max. performance	Yes
(b) One-engine-inoperative procedures (multiengine)	Yes
ii) Cruise	Yes
(a) Performance characteristics (speed vs. power)	Yes
(b) Normal and steep turns	Yes
(c) Approach to stalls (i.e., stall warning), stalls, and recovery. Execute from takeoff, cruise, and approach and landing configurations.	Yes
(d) In-flight engine shutdown (multiengine)	Yes
(e) Fuel selector function	Yes
(f) In-flight engine start	Yes
iii) Approach	Yes
(a) Normal (with and without flaps) (check gear warning, if applicable)	Yes
(b) Best glide no power	Yes
iv) Landings	Yes
2) HELICOPTER In-Flight Operation	
i) Hovering and air taxi	NA

(a) Forward	NA
(b) Rearward	NA
(c) Sideward	NA
(d) Turns	NA
ii) Climb	NA
iii) Cruise	NA
(a) Performance characteristics (speed vs. power)	NA
Maneuvers and Tasks	**Yes/No/NA**
(b) Turns	NA
(i) Recovery	NA
(ii) Skidding	NA
(iii) Slipping	NA
(iv) Steep turns	NA
(c) In-flight engine shutdown and start (multiengine)	NA
(d) Descents	NA
(e) Straight in and 180º autorotation	NA
(f) Landings	NA
d) Instrument Approaches	
1) Nonprecision	
i) GPS and LPV	Yes
ii) GPS-WAAS (optional)	Varies* (Based on Configuration)
iii) All engines operating	Yes
iv) One or more engines inoperative	Yes
v) Approach procedures (VOR, VOR/DME, LOC procedures on an ILS, LDA, RNAV (RDP) or RNAV (GPS) to LNAV, LNAV/VNAV or LPV)	Yes
2) Precision	
i) ILS	Yes
ii) GLS (optional)	No
iii) Effects of crosswind	Yes
iv) With engine inoperative (multiengine)	Yes
v) Missed approach	Yes
(a) Normal	Yes
(b) With engine(s) inoperative (multiengine)	Yes
e) Surface Operations	
1) AIRPLANE Surface Operations (Post Landing)	
i) Approach and landing roll	Yes
ii) Braking operation	Yes
Maneuvers and Tasks	**Yes/No/NA**
iii) Reverse thrust operation, if applicable	NA
2) HELICOPTER Surface Operations	
i) Landings	NA
ii) Landing area operations	NA

f) HELICOPTER Emergency Operations	
1) Power failure at hover	NA
2) Power failure at altitude	NA
3) System and equipment malfunctions	NA
4) Settling with power (optional)	NA
5) Low rotor RPM recovery (optional)	NA
6) Antitorque system failure	NA
7) Dynamic rollover (optional)	NA
g) Any Flight Phase	
1) Aircraft and Powerplant Systems	Yes
i) Electrical, mechanical, or hydraulic	Yes
ii) Flaps (airplane)	Yes
iii) Fuel selector and oil temp/pressure	Yes
iv) Landing gear (if applicable)	Yes
2) Flight Management and Guidance Systems	Yes
i) Autopilot (if standard equipment)	Yes
ii) Flight director (AATD only)/system displays (if installed)	Yes
iii) Navigation systems	Yes
iv) Stall warning systems avoidance (airplane)	Yes
v) Multi-function displays (if applicable)	Varies* (Based on Configuration)
3) Airborne Procedures	Yes
i) Holding	Yes
ii) Uncoordinated turns – slipping and skidding demo	Yes
Maneuvers and Tasks	**Yes/No/NA**
iii) Configuration and power changes and resulting pitch changes	Yes
iv) Compass turns and appropriate errors (if installed)	Yes
4) Engine Shutdown and Parking	Yes
i) Systems operation	Yes
ii) Parking brake operation (if installed) (airplane)	Yes
h) Can simulate engine failure, including failures due to simulated loss of oil pressure or fuel starvation.	Yes (Except oil pressure loss)
i) Can simulate the following equipment or system failures:	
1) Alternator or generator failure.	Yes
2) Vacuum pump/pressure failure and the associated flight instrument failures.	Yes
3) Gyroscopic flight instrument failures.	Yes
4) Pitot/static system malfunction and the associated flight instrument failures.	Yes
5) Electronic flight deck display malfunctions.	Yes
6) Landing gear (if retractable) or flap malfunctions.	Yes

j) Independent Instructor Station Requirements (AATD Only)	
1) Displays published airways and holding patterns.	Yes
2) Displays aircraft position and track.	Yes
3) Displays aircraft altitude and speed.	Yes
4) Displays NAVAIDs and airports.	Yes
5) Can record and replay aircraft ground track history for entire training session.	Yes
6) Can invoke instrument or equipment failures.	Yes

Any changes or modifications to this device which have not been reviewed, evaluated, and approved in writing by General Aviation and Commercial Division, AFS-800 may terminate FAA approval of this aviation training device.